

# **Overcoming the Split Incentive in Multi-Family Urban Housing: An Integrated Earth-Observation and Tariffed On-Bill Electrification Framework**

## **Decarbonization and Resilience Imperatives in the Built Environment**

The global climate framework established under COP31 sets ambitious decarbonization benchmarks: achieving 35% electrification of final global energy consumption and reducing building-sector energy intensity by 25% by 2035<sup>1</sup>. The building sector accounts for over one-third of global final energy consumption and nearly 38% of global energy-related greenhouse gas emissions when accounting for upstream materials and construction<sup>4</sup>. Operational energy within residential structures is dominated by thermal conditioning, where space heating, space cooling, and domestic water heating account for 35% to 60% of total building energy demand<sup>6</sup>. Consequently, transitioning multi-family residential housing from fossil-fuel hydronic and furnace systems to high-efficiency electric heat pumps constitutes a critical pathway for rapid emissions abatement<sup>7</sup>.

This mitigation imperative is deeply complicated by compounding climate extremes, specifically the amplification of heatwaves across dense urban centers<sup>9</sup>. The Urban Heat Island (UHI) effect elevates canopy-layer ambient temperatures by 2°C to 8°C relative to adjacent rural baselines, driven by dense building geometries, low-albedo building envelopes, and the retention of sensible anthropogenic heat<sup>9</sup>. Elevated urban temperatures increase cooling energy consumption by 10% to 50% in multi-family dwellings, straining urban electrical distribution grids during summer peak hours and accelerating equipment degradation<sup>11</sup>. Without coordinated intervention, the global installed capacity of cooling equipment is projected to expand from 22 TW to 68 TW by 2050, triggering severe grid infrastructure bottlenecks and offsetting efficiency gains<sup>14</sup>.

Meeting the COP31 targets necessitates an integrated design framework that treats structural electrification, passive thermal envelope resilience, and localized urban microclimate dynamics as interconnected components of a unified urban system<sup>7</sup>.

## **The Split-Incentive Impasse and Urban Thermal**

## Disparities

Progress toward electrifying and weatherizing urban residential structures is largely obstructed by a foundational institutional failure: the *split incentive*<sup>17</sup>. This principal-agent misalignment is acute within tenant-occupied multi-family residential properties, where the building owner is legally responsible for capital expenditures (CapEx) such as roof replacements, heating, ventilation, and air conditioning (HVAC) infrastructure, and structural insulation<sup>12</sup>. However, the recurring operational expenditure (OpEx) benefits resulting from these investments—lower monthly utility bills and improved thermal comfort—accrue exclusively to tenants who pay their own electric and natural gas bills<sup>18</sup>. Landlords have no direct financial mechanism to monetize these utility reductions and are therefore disincentivized from incurring capital debt for high-performance upgrades<sup>19</sup>. Tenants, meanwhile, lack both property equity and legal authority to alter building envelopes or install central heat pump systems, leaving both parties unable to act<sup>24</sup>.

This impasse disproportionately burdens low-to-moderate-income (LMI) residents living in older, poorly insulated multi-family properties<sup>18</sup>. These properties are often concentrated in dense, tree-deficient urban zones where degraded roofing assemblies absorb excessive solar radiation, causing interior overheating during summer extremes<sup>11</sup>. Low-income households face elevated energy burdens, paying disproportionate shares of their income toward inefficient cooling and heating systems while lacking the capital or credit required for conventional private financing mechanisms<sup>18</sup>.

Conventional clean energy financing mechanisms—such as personal debt, commercial property loans, and Property Assessed Clean Energy (PACE) financing—fail to resolve the multi-family split incentive<sup>25</sup>. Debt-based instruments encounter restrictive underwriting criteria, risk debt aversion from property owners, and remain tied to individual credit ratings rather than physical building performance<sup>25</sup>. In affordable housing, strict regulatory rent ceilings prevent landlords from passing capital improvement costs through to tenant rents, reinforcing deferred maintenance and leaving tenants exposed to extreme heat and volatile fossil-fuel prices<sup>20</sup>.

| Housing Tenure Segment       | Primary Financial Constraint                                       | Principal-Agent Dynamic                                                                                                       | Energy & Resilience Deficit                                                                                               |
|------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Owner-Occupied Single Family | High upfront capital cost and personal debt limits <sup>26</sup> . | Direct alignment: Capital investor directly captures operational utility savings and property valuation gains <sup>28</sup> . | Moderate: Constrained by personal liquidity and contractor availability rather than misaligned incentives <sup>18</sup> . |

|                                                        |                                                                                                                    |                                                                                                                                              |                                                                                                                              |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Master-Metered Multi-Family</b>                     | Capital allocation constraints and lack of individual tenant sub-metering <sup>26</sup> .                          | Partial alignment: Landlord captures operational savings directly, but tenants face zero marginal cost for excess energy use <sup>26</sup> . | High: Minimal incentive for tenant behavioral conservation; equipment is run continuously under peak loads <sup>24</sup> .   |
| <b>Individually Metered Multi-Family (Tenant-Paid)</b> | Structural split incentive; capital costs fall on owners while operating savings accrue to tenants <sup>18</sup> . | Total misalignment: Property owners cannot monetize utility bill reductions; tenants lack capital and property rights <sup>19</sup> .        | Severe: Systemic underinvestment in heat pump conversions, envelope insulation, and cool roofing <sup>18</sup> .             |
| <b>Subsidized &amp; Social Housing</b>                 | Constrained municipal budgets and statutory limitations on tenant rent escalation <sup>20</sup> .                  | Administrative deadlock: Housing operators must balance capital maintenance backlogs against decarbonization mandates <sup>24</sup> .        | Extreme: Highly vulnerable to extreme heat events; reliance on inefficient resistive electric or gas heating <sup>18</sup> . |

Overcoming this split incentive requires an institutional and technical framework that decouples capital recovery from both tenant personal credit and landlord balance-sheet liability, transforming building retrofits into utility-grade, meter-attached infrastructure investments<sup>19</sup>.

## Architectural Design of the Geospatial-Thermal On-Bill Engine

To resolve the split-incentive impasse, an automated socio-technical framework is formulated: the *Geospatial Urban Building Energy and Tariffed Optimization* (GUBETO) system. The platform integrates open-access Earth Observation (EO) data, physics-based Urban Building Energy Modeling (UBEM), and an inclusive utility-administered Pay As You Save (PAYS) tariff model<sup>9</sup>. GUBETO establishes a standardized pathway for municipal housing authorities, electric utilities, and building owners to evaluate, finance, and execute multi-family decarbonization retrofits without upfront capital expenditure or personal debt obligations<sup>25</sup>.

The platform operates by executing a continuous computational and financial sequence across three core modules:

The first operational stage relies on the spatial microclimate engine. Regional thermodynamic forcing data derived from the Copernicus Climate Change Service (C3S) ERA5-Land reanalysis dataset are downscaled from their native 9-kilometer resolution using multispectral and thermal infrared imagery from Sentinel-2, Sentinel-3, and Landsat 8/9<sup>9</sup>. By integrating the Copernicus Land Monitoring Service (CLMS) Percentage Impervious Surface (PIS) metrics, the module maps local land surface temperatures and canopy-layer heat island intensities at a 100-meter horizontal resolution<sup>9</sup>. This microclimate downscaling step produces localized EnergyPlus Weather (EPW) data streams that accurately reflect localized heat stress rather than generic, airport-derived weather baselines<sup>13</sup>.

The second stage feeds the downscaled microclimate time series into an automated urban building energy simulation framework powered by the open-source URBANopt Software Development Kit (SDK) and OpenStudio-HPXML<sup>15</sup>. Municipal cadastral data and footprint geometries are ingested via open GeoJSON feature files to construct baseline multi-family building energy models<sup>33</sup>. The simulation evaluates two paired interventions: a passive thermal envelope upgrade deploying high-reflectance cool roofs meeting ASTM E1980 criteria (aged Solar Reflectance Index  $SRI \geq$  ), followed by an active mechanical upgrade replacing fossil-fuel or resistive heating with variable-refrigerant-flow (VRF) cold-climate air-source heat pumps<sup>5</sup>.

The third stage passes the simulated hourly energy differentials into an algorithmic Pay As You Save (PAYS) tariff clearance engine<sup>25</sup>. The software evaluates whether the combined passive-envelope and active-electrification retrofit package satisfies strict utility bill neutrality for the tenant<sup>19</sup>. Once validated, the engine formats the financial schedule into a location-specific utility tariff rider tied directly to the building meter, enabling immediate execution by municipal program managers and contracted installation teams<sup>21</sup>.

## Microclimate Downscaling and Environmental Modeling

The environmental modeling engine processes atmospheric parameters including 2-meter air temperature, surface pressure, dewpoint temperature, and downward shortwave and longwave radiation fluxes from the Copernicus ERA5-Land reanalysis dataset<sup>34</sup>. Native 9-kilometer grid cells cannot resolve intra-urban microclimates shaped by building geometry, aspect ratio, canyon orientation, and surface materials<sup>10</sup>. GUBETO overcomes this limitation by implementing a dynamic split-window thermal downscaling methodology<sup>9</sup>.

Thermal infrared radiance captured by Landsat 8/9 TIRS and Sentinel-3 is combined with fractional vegetation cover ( $F_v$ ) derived from Sentinel-2 Normalized Difference Vegetation Index (NDVI) values to calculate surface emissivity ( $\epsilon$ )<sup>9</sup>. The surface energy balance across each 100-meter urban grid cell is calculated by resolving net surface radiation ( $R_n$ ), sensible anthropogenic heat emissions ( $Q_F$ ), sensible turbulent heat flux ( $H$ ), latent turbulent heat

flux ( $LE$ ), and conductive substrate heat storage flux ( $G$ )<sup>41</sup>:

$$R_n + Q_F = H + LE + G$$

Sensible anthropogenic heat flux ( $Q_F$ ) incorporates vehicular, mechanical HVAC exhaust, and metabolic heat re-radiation<sup>11</sup>. Anthropogenic emissions and impervious surface densities derived from CLMS regional layers are evaluated through kernel density estimation, generating localized Canopy Layer Heat Island (CLHI) anomalies<sup>9</sup>. These anomalies are superimposed onto the ERA5-Land baseline, generating microclimate-adjusted hourly EPW files that capture neighborhood-specific extreme heat events<sup>13</sup>.

## Automated Urban Building Energy Simulation and Envelope Sizing

The generated microclimate files drive district-scale energy simulations executed within the URBANopt SDK via Ruby-based OpenStudio Measure scripts<sup>15</sup>. Urban parcels parsed from municipal GIS databases provide structural characteristics, gross floor areas, and window-to-wall ratios<sup>33</sup>. Using these geometric parameters, the engine models the baseline performance of multi-family buildings and assesses the impact of two coordinated interventions<sup>33</sup>.

The passive envelope intervention models the application of cool roof technology conforming to ASTM E1980<sup>12</sup>. Dark built-up or asphalt membrane roofs with an initial Solar Reflectance

Index ( $SRI$ ) between 6 and 15 absorb up to 90% of incident solar irradiance, elevating rooftop surface temperatures to 65°C–70°C and driving severe conductive heat flux into top-floor living units<sup>12</sup>. The engine models the application of elastomeric acrylic coatings or thermoplastic polyolefin (TPO) single-ply membranes engineered with an aged solar

reflectance  $\rho \geq 0.70$  and thermal emittance  $\epsilon \geq 0.85$ , achieving an aged  $SRI \geq$  <sup>12</sup>.

This passive intervention lowers external surface temperatures by 10°C to 20°C and reduces building-level cooling loads by 10% to 40%<sup>12</sup>.

The active mechanical intervention replaces inefficient space conditioning—such as natural gas hydronic boilers or electric baseboards paired with legacy window air conditioners—with centralized or multi-split variable-speed air-source heat pumps<sup>5</sup>. Because the cool roof deployment significantly dampens peak solar heat gain, the active mechanical system can be right-sized<sup>12</sup>. Right-sizing the heat pump lowers equipment capital expenditure, avoids costly electrical service panel upgrades, and mitigates distribution grid stress during regional peak hours<sup>16</sup>.

## Algorithmic Pay As You Save Tariff Integration

To bypass the split-incentive impasse, GUBETO applies the Pay As You Save (PAYS) system, an established financing framework for tariffed on-bill investments<sup>25</sup>. Under PAYS, capital improvements are not structured as loans, debt obligations, or personal financial liabilities for

either the tenant or the property owner<sup>25</sup>. Instead, the utility deploys capital to install site-specific, cost-effective energy infrastructure behind the meter, treating the upgrade on par with traditional utility system investments<sup>19</sup>.

Cost recovery is achieved through a fixed, recurring tariff charge assigned to the physical utility meter<sup>19</sup>. The tariff structure enforces strict *bill neutrality*, guaranteeing that the total post-retrofit monthly expenditure (the new volumetric energy charge plus the tariff cost-recovery charge) does not exceed the pre-retrofit baseline bill<sup>19</sup>:

$$\text{Post-Retrofit Energy Charge}_m + \text{Tariff Charge}_m \leq \alpha \times \text{Pre-Retrofit Baseline Charge}_m$$

where  $m$  represents the billing month, and  $\alpha$  represents the statutory program coverage factor, set to **0.85** to guarantee a minimum 15% net reduction in total monthly energy expenses for participating tenants<sup>25</sup>.

The maximum allowable capital expenditure ( $\text{CapEx}_{\max}$ ) for an eligible retrofit package is calculated dynamically based on simulated annual utility cost savings ( $\Delta S_{\text{annual}}$ ), the utility weighted average cost of capital ( $r$ ), an amortization horizon ( $n \leq 15$  years), and a 5% contingency reserve against bad debt or equipment performance variations<sup>25</sup>:

$$\text{CapEx}_{\max} = (1 - \text{Reserve}) \times \sum_{t=1}^n \frac{\alpha \times \Delta S_{\text{annual}}}{(1 + r)^t}$$

Because the tariff obligation is attached directly to the meter, it automatically transfers to subsequent tenants or property owners upon occupancy changes, eliminating debt burdens for short-term occupants and removing landlord balance-sheet liability<sup>19</sup>.

| System Component                         | Technology & Data Foundations                                             | Core Functional Operation                                                                 | Direct Decarbonization Mechanism                                                          |
|------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Microclimate Downscaling Pipeline</b> | Copernicus ERA5-Land, Sentinel-2/3, Landsat-8/9 TIRS, CLMS <sup>9</sup> . | Downscales 9 km regional climate data to 100 m neighborhood thermal fields <sup>9</sup> . | Replaces rural airport weather baselines with localized microclimate data <sup>13</sup> . |
| <b>District Energy Modeling Module</b>   | URBANopt SDK, OpenStudio-HPXML, EnergyPlus Engine <sup>33</sup> .         | Translates GIS polygons into multi-zone thermal simulation                                | Quantifies hourly savings from envelope retrofits and equipment                           |

|                                        |                                                                                     |                                                                                                     |                                                                                                |
|----------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|                                        |                                                                                     | archetypes <sup>15</sup> .                                                                          | electrification <sup>16</sup> .                                                                |
| <b>Passive Envelope Interventions</b>  | High-reflectance cool roofs (ASTM E1980, aged $SRI \geq \quad$ ) <sup>12</sup> .    | Reflects shortwave solar radiation and maximizes thermal infrared emittance <sup>12</sup> .         | Reduces cooling loads by 10% to 40% and lowers rooftop surface temperatures <sup>12</sup> .    |
| <b>Active Electrification Systems</b>  | Variable-speed cold-climate air-source heat pumps (ASHP) <sup>39</sup> .            | Replaces combustion and resistive heating with vapor-compression cycles <sup>5</sup> .              | Eliminates onsite natural gas combustion and lowers seasonal thermal energy use <sup>3</sup> . |
| <b>PAYS Tariff Optimization Engine</b> | Algorithmic tariff parser integrated with utility billing schedules <sup>25</sup> . | Restricts CapEx to levels supported by verified bill neutrality ( $\alpha = 0.85$ ) <sup>19</sup> . | Resolves split incentive; attaches recovery to meter without tenant debt <sup>21</sup> .       |

## Computational Prototyping and Quantitative Validation

The technical viability and economic mechanics of the GUBETO framework are demonstrated using a computational prototype applied to a representative urban testbed. The testbed comprises 12 masonry multi-family residential buildings constructed between 1960 and 1980, encompassing 240 tenant-occupied units located within a recognized urban heat island corridor<sup>32</sup>.

The computational prototype executes an automated sequence across the testbed parcels. First, the microclimate module extracts hourly ERA5-Land reanalysis fields for the peak summer months of June through August, coupling them with Landsat 8 TIRS thermal band acquisitions and Sentinel-2 NDVI layers<sup>9</sup>. The downscaled canopy-layer temperature fields identify a localized daytime temperature anomaly of +3.8°C above the regional baseline recorded at the municipal airport weather station<sup>9</sup>. These localized adjustments are compiled into a microclimate-corrected EPW weather file<sup>13</sup>.

Second, the URBANopt simulation pipeline establishes baseline building operations using OpenStudio-HPXML archetypes<sup>33</sup>. The baseline configuration models atmospheric natural gas boilers with a seasonal heating efficiency ( $\eta$ ) of 0.72, legacy window air conditioners with an

Energy Efficiency Ratio (**EER**) of 9.0 ( $2.64 W_{th}/W_e$ ), and an aged built-up asphalt roof membrane with a solar reflectance of 0.12 and an **SRI** of 12<sup>5</sup>. The intervention scenario applies a high-performance elastomeric cool roof membrane with an aged **SRI** of 88 ( $\rho =$  ,  $\epsilon = 0.90$ ) combined with cold-climate variable-speed air-source heat pumps achieving a seasonal coefficient of performance (**COP**) of 3.2 at -8°C and a Seasonal Energy Efficiency Ratio (**SEER**) of 18.0 ( $5.27 W_{th}/W_e$ )<sup>39</sup>. The simulation results confirm that passive envelope cooling significantly alters mechanical heat pump sizing requirements<sup>12</sup>. Across the 240 units, the cool roof reduces peak rooftop surface temperatures during July extremes from 68.4°C down to 41.2°C, delivering a 27.2°C surface temperature reduction that decreases top-floor peak cooling thermal loads by 34.6%<sup>12</sup>. Consequently, the required cooling capacity for the heat pump systems is reduced from 4.2 kW (1.20 tons) per unit down to 2.3 kW (0.65 tons) per unit<sup>39</sup>. This 45.2% reduction in peak mechanical cooling capacity lowers upfront equipment and installation costs by 28%, eliminating the need for expensive main electrical panel upgrades across the housing cluster<sup>16</sup>. The combined intervention package eliminates onsite fossil gas consumption, reducing direct combustion emissions by 100% while cutting total annual site Energy Use Intensity (EUI) by 52.7%, from 194.2 to 91.8  $kWh/m^2/year$ <sup>3</sup>. When passed to the financial optimization engine, the combined utility cost reductions from heating, cooling, and water heating generate an average pre-retrofit monthly baseline savings of \$108.00 per unit<sup>25</sup>.

Assuming a utility weighted average cost of capital ( $r$ ) of 4.5% and a 15-year recovery horizon, the maximum allowable PAYS capital expenditure is calculated at \$10,480 per residential unit<sup>25</sup>. The actual combined turnkey installation cost for the cool roof membrane (1,650 per unit share) and the right-sized heat pump system (7,800 per unit) totals \$9,450 per unit, falling fully within the allowable capital threshold<sup>12</sup>. Under the resulting PAYS tariff, the utility recovers capital costs through a fixed monthly meter charge of \$81.50 per unit<sup>25</sup>. The tenant's post-retrofit electricity charge averages \$54.00, resulting in a total net monthly energy outlay of \$135.50<sup>25</sup>. Compared to the pre-retrofit baseline expenditure of

<sup>25</sup>.  
162.00 per month (combined fossil gas and electricity charges), the tenant achieves an immediate, guaranteed net cost savings of \$26.50 per month, or 16.4% ( $\alpha = 0.836$ )

The split incentive is fully bypassed: the landlord incurs zero upfront debt and no balance-sheet liability, while the tenant captures lower monthly energy expenses and improved indoor thermal comfort without personal credit underwriting<sup>18</sup>.

| Performance Dimension               | Baseline Profile                                    | GUBETO Retrofit Package                            | Absolute & Percentage Delta                    |
|-------------------------------------|-----------------------------------------------------|----------------------------------------------------|------------------------------------------------|
| Roof Membrane Albedo / SRI          | $\rho =$ ,<br>SRI =<br>[cite: 40]                   | $\rho =$ ,<br>SRI =<br>[cite: 40]                  | +76 SRI points (-<br>[cite: 40])               |
| Peak Summer Roof Surface Temp       | 68.4°C<br>[cite: 12]                                | 41.2°C<br>[cite: 12]                               | -27.2°C (-39.8%<br>[cite: 12])                 |
| Site Energy Use Intensity (EUI)     | 194.2 kWh/m <sup>2</sup> /y.<br>[cite: 5]           | 91.8 kWh/m <sup>2</sup> /y.<br>[cite: 5]           | -102.4 kWh/m <sup>2</sup> /<br>[cite: 5])      |
| Onsite Fossil Gas Consumption       | 48.2 MJ/m <sup>2</sup> /y <sub>1</sub><br>[cite: 5] | 0.0 MJ/m <sup>2</sup> /y <sub>1</sub><br>[cite: 5] | -48.2 MJ/m <sup>2</sup> /yr<br>[cite: 5])      |
| Peak Summer Mechanical Cooling Load | 4.2 kW/unit<br>[cite: 39]                           | 2.3 kW/unit<br>[cite: 39]                          | -1.9 kW/unit (-<br>[cite: 39])                 |
| Monthly Tenant Energy Burden        | \$162.00/month<br>[cite: 25]                        | \$135.50/month<br>(incl. tariff) <sup>25</sup>     | -\$26.50/month (-<br>[cite: 25])               |
| Utility Capital Amortization Term   | Not Applicable                                      | 15 years at 4.5%<br>[cite: 25]                     | Fully amortized via meter tariff <sup>19</sup> |

## Real-World Implementation Pathway and Institutional Mechanics

Translating the GUBETO framework into municipal practice requires establishing clear institutional coordination, regulatory frameworks, and phased execution timelines<sup>25</sup>.

The implementation roadmap spans four sequential phases over a 24-month horizon: During Phase 1 (Months 1–6: Architecture Calibration and Integration), technical working groups integrate municipal GIS cadastral layers with automated data pipelines connected to the Copernicus Climate Data Store API and Landsat thermal repositories<sup>32</sup>. Downscaled microclimate models and automated URBANopt script templates are calibrated against municipal weather networks to ensure local accuracy<sup>9</sup>.

Phase 2 (Months 7–12: Regulatory Structuring and Utility Tariff Filing) focuses on legal and

financial alignment. The municipal administration, participating electric utilities, and regulatory bodies draft and submit a dedicated PAYS tariff rider to the state or national public utility commission<sup>25</sup>. The rider formally establishes the utility's authority to capitalize behind-the-meter building retrofits, defines the non-debt, location-specific cost recovery mechanism, and sets statutory consumer protections ensuring guaranteed bill neutrality ( $\alpha \leq 0.85$ )<sup>19</sup>. A financing facility backed by municipal green bonds or public green banks is established to provide low-cost capital to the utility<sup>4</sup>.

Phase 3 (Months 13–18: Urban Pilot Deployment and Sensor Validation) executes physical installations across a representative pilot portfolio of 10 multi-family buildings (approximately 200 residential units)<sup>26</sup>. Certified local mechanical and roofing contractors install high-SRI cool roof membranes and right-sized variable-speed heat pumps<sup>26</sup>. High-resolution IoT electrical sub-meters and wireless indoor temperature/humidity loggers are installed to track energy use and verify indoor thermal conditions<sup>37</sup>.

Phase 4 (Months 19–24: District-Scale Rollout and Algorithmic Automation) transitions the system from pilot validation to automated district scaling. A public web-based platform is deployed, allowing housing authorities, non-profit community housing providers, and private landlords to automatically generate preliminary building audits, calculate PAYS investment parameters, and connect with qualified installation contractors<sup>37</sup>. Programmatic bulk purchasing of heat pumps and roofing materials is implemented to capture supply chain economies of scale, lowering per-unit capital costs across municipal social housing portfolios<sup>26</sup>.

| Operational Phase                                   | Strategic Objective                                          | Primary Technical & Regulatory Deliverables                                                                              | Leading Entities                                                                                 |
|-----------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Phase 1: Architecture Calibration</b><br>(M1–M6) | Multi-scale model calibration and GIS cadastral integration. | Integrated open-source software pipeline producing microclimate EPWs and URBANopt models <sup>9</sup> .                  | Municipal GIS departments, Academic research partners, Energy modeling engineers <sup>33</sup> . |
| <b>Phase 2: Regulatory Structuring</b><br>(M7–M12)  | Public utility commission (PUC) tariff approval.             | Approved PAYS tariff docket establishing meter-attached capital recovery with guaranteed bill neutrality <sup>19</sup> . | Electric utilities, Public utility commissions, Affordable housing coalitions <sup>25</sup> .    |

|                                                     |                                                         |                                                                                                                           |                                                                                                                         |
|-----------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 3: Urban Pilot Deployment</b><br>(M13–M18) | Empirical pilot retrofitting of 10 multi-family assets. | Physical deployment of high-SRI roofs and heat pumps; sensor validation of energy and thermal comfort <sup>12</sup> .     | Certified retrofit contractors, Housing authority operations teams, Measurement & verification agents <sup>26</sup> .   |
| <b>Phase 4: District-Scale Rollout</b> (M19–M24)    | Programmatic automated rollout across urban blocks.     | Multi-tenant web platform generating automated feasibility audits, contractor bids, and on-bill contracts <sup>37</sup> . | Municipal housing authorities, Commercial energy service providers, Utility customer service operations <sup>25</sup> . |

Multi-stakeholder collaboration is central to system operations. Municipal housing authorities manage asset selection and execute building-access agreements, securing property owner consent through an investment proposition that requires no corporate debt and creates no property liens<sup>25</sup>. Participating electric utilities deploy project capital, track ongoing performance, and incorporate cost-recovery charges into their customer billing systems<sup>21</sup>. Regional contractor networks deliver retrofits according to performance-based installation standards, ensuring that mechanical equipment matches the reduced thermal sizing requirements enabled by cool roof installations<sup>12</sup>.

Long-term project integrity requires addressing specific operational and environmental risks. Atmospheric particulate deposition, biological accumulation, and weathering degrade roof reflectance over time, lowering initial SRI values by 15% to 20% over three years<sup>12</sup>. To preserve project economics, the GUBETO simulation engine mandates the use of 3-year aged SRI values

(minimum aged  $SRI \geq 64$  per LEED v4.1 and ASHRAE 90.1 standards) rather than initial unaged ratings, preventing the overestimation of long-term cooling savings<sup>40</sup>. Periodic surface washing protocols are integrated into utility maintenance contracts to maintain reflective performance<sup>12</sup>.

Customer protection safeguards are essential when working with low-income populations. Under PAYS program guidelines, meter-attached charges must not increase the risk of utility shut-offs for participating households<sup>25</sup>. Because the system enforces strict bill neutrality (

$\alpha = 0.85$ ), participating tenants experience an immediate net reduction in their monthly bills, lowering default risk<sup>19</sup>. A program-level Loan Loss Reserve funded by federal and municipal climate grants covers temporary payment arrears, protecting the utility capital facility without triggering residential disconnections<sup>25</sup>.

Finally, equipment performance risks are addressed through contractual warranty requirements. If an installed heat pump experiences mechanical failure or operational

underperformance, tariff collection on the tenant's bill is paused until the contractor resolves the issue under mandatory multi-year service warranties, aligning contractor incentives directly with operational reliability<sup>19</sup>.

## Strategic Synthesis and Systemic Implications

The GUBETO framework addresses the multi-family building decarbonization challenge by transforming what is typically treated as a legal and financial impasse into an integrated computational and tariffing workflow<sup>17</sup>. By combining satellite-derived urban microclimate data with physics-based building modeling, the system eliminates administrative overhead and sizing errors that historically lead to oversized heat pumps and failed financial models<sup>9</sup>.

Integrating passive cool-roof interventions with active heat pump electrification alters the economic mechanics of residential decarbonization<sup>7</sup>. Lowering the building envelope's solar heat gain reduces peak cooling loads by up to 45%, protecting electrical distribution infrastructure from transformer overloads during extreme heatwaves while keeping capital costs within bounds that can be fully amortized through utility bill savings<sup>16</sup>.

By utilizing utility-administered Pay As You Save tariffs, the framework eliminates the split-incentive barrier in rental housing<sup>21</sup>. Landlords avoid upfront capital debt, while tenants secure lower energy costs and protection against urban heatwaves<sup>18</sup>.

This integrated framework offers a scalable, replicable pathway for cities worldwide to modernize aging multi-family housing, protect vulnerable communities from intensifying climate risks, and meet COP31 electrification and energy-reduction targets by 2035<sup>2</sup>.

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